

# Applied LLMs for Forecasting and Policy Research

## Lecture 1: Large Language Models

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# Reading

Background material for the lecture: Jurafsky and Martin, *Speech and Language Processing*, <https://stanford.io/4qkv65V>, Ch. 7, 8, 10.

# Outline

Introduction

Basic Concepts

Attention and Transformer Models

Economic Applications

# Motivation for Mini-Course

Text-as-data methods have been used for years in economics and finance [Gentzkow et al., 2019a, Ash and Hansen, 2023].

Also, growing use of image data [Borgschulte et al., 2021, Ash et al., 2022, Adukia et al., 2023].

Unstructured datasets are primarily used to **measure** otherwise hard-to-quantify economic variables:

1. Sentiment [Shapiro et al., 2022]
2. Innovation [Kelly et al., 2021]
3. Skill demand [Deming and Kahn, 2018]
4. Firm-level macro shock exposures [Hassan et al., 2019, Davis et al., 2020]
5. Occupation exposure to technical change [Kogan et al., 2019]

# What Do Large Language Models Add?

The traditional way of representing text data is the **bag-of-words** model where each document is a histogram count over vocabulary terms.

Produced influential work such as [Gentzkow and Shapiro, 2010, Baker et al., 2016, Gentzkow et al., 2019b].

But such methods struggle to account for how sequences of words produce meaning.

# Limitations of Bag-of-Words

## Synonymy

*economic growth is weak but long-term productivity trends are strong*  
*economic growth is tepid but long-term productivity trends are strong*

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*my cat's favorite activity is to **lie** on our bed*

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## Sequence

*economic growth is weak but long-term productivity trends are strong*  
*economic growth is strong but long-term productivity trends are weak*



# Language as Vectors

At a very high level, an LLM takes an input text and converts it into a (relatively) low dimensional vector.

This vector might be of interest in its own right, e.g. document similarity.

Or it might be an intermediate representation for a different language task, e.g. predicting an associated label or predicting new words.

LLMs allow the vector representation to change when the word sequence changes to capture meaning.

NB: this added flexibility may or may not dramatically improve the performance of the model for economic measurement tasks.

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# Notation

The corpus is composed of  $D$  documents indexed by  $d$ .

After pre-processing, each document is a finite, length- $N_d$  list of terms  $\mathbf{w}_d = (w_{d,1}, \dots, w_{d,N_d})$  with generic element  $w_{d,n}$ .

$N \equiv \sum_d N_d$  is the total number of terms in the corpus.

Suppose there are  $V$  **unique** terms in the corpus, where  $1 \leq V \leq N$ , each indexed by  $v$ .

We can then map each term in the corpus into this index, so that  $w_{d,n} \in \{1, \dots, V\}$ .

# Example

Consider three documents:

1. 'stephen is nice'
2. 'john is also nice'
3. 'george is mean'

We can consider the set of unique terms as {stephen, is, nice, john, also, george, mean} so that  $V = 7$ .

Construct the following index:

| stephen | is | nice | john | also | george | mean |
|---------|----|------|------|------|--------|------|
| 1       | 2  | 3    | 4    | 5    | 6      | 7    |

We then have  $\mathbf{w}_1 = (1, 2, 3)$ ;  $\mathbf{w}_2 = (4, 2, 5, 3)$ ;  $\mathbf{w}_3 = (6, 2, 7)$ .

# Word Embeddings

A **word embedding** is a low-dimensional vector representation of a word.

Ideally in this low-dimensional vector space words with similar meanings will lie close together.

The construction of word embeddings was an important precursor to the development of large language models.

# Distributional Hypothesis

The **distributional hypothesis** states that words that share similar contexts share similar meanings.

Example from Jurafsky and Martin:

(6.1) Ongchoi is delicious sauteed with garlic.

(6.2) Ongchoi is superb over rice.

(6.3) ...ongchoi leaves with salty sauces...

And suppose that you had seen many of these context words in other contexts:

(6.4) ...spinach sauteed with garlic over rice...

(6.5) ...chard stems and leaves are delicious...

(6.6) ...collard greens and other salty leafy greens

# Formalizing Local Context

Recall that  $w_{d,n}$  is the  $n$ th word in document  $d$ .

The *context* of  $w_{d,n}$  is a length- $2L$  window of words around  $w_{d,n}$ :

$$C(w_{d,n}) = [w_{d,n-L}, w_{d,n-L+1}, \dots, w_{d,n+L-1}, w_{d,n+L}]$$

Can truncate context appropriately if window stretches past beginning or end of text.

In line with distributional hypothesis, word embedding models seek to generate similar embeddings for words that share similar contexts.

# Word2Vec

**Word2vec** [Mikolov et al., 2013a, Mikolov et al., 2013b] is a particularly well-known algorithm for the construction of word embeddings.

Important example of a neural-network-based language model that was scalable and effective.



# Self-Supervised Learning

The ‘meaning’ of a word is an unobserved and subjective concept.

Difficult to directly formulate an objective function.

Important conceptual idea is to formulate word prediction tasks that are solved using word embeddings.<sup>1</sup>

The approach of using auxiliary word prediction tasks to build high-quality embeddings is called [self-supervised learning](#).

## Example: skipgram variant of Word2vec

1. Predict **presence** of each  $w_{d,n-l} \in C(w_{d,n})$  given  $w_{d,n}$ .
2. Predict **absence** of randomly sampled words from the corpus given  $w_{d,n}$ .

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<sup>1</sup>See also [Bengio et al., 2003].

# How Close are Words?

The standard way of judging whether word vectors are “close” is **cosine similarity**.

$$CS(i, j) = \frac{\rho_i \cdot \rho_j}{\|\rho_i\| \|\rho_j\|}$$

Measures whether vectors point in the same direction.

# Terms Close to Uncertainty in FOMC Transcripts

| term          | sim   | term         | sim   |
|---------------|-------|--------------|-------|
| uncertainties | 0.741 | challenges   | 0.415 |
| anxiety       | 0.48  | fragility    | 0.405 |
| pessimism     | 0.479 | clarity      | 0.401 |
| skepticism    | 0.465 | concerns     | 0.4   |
| optimism      | 0.445 | risks        | 0.397 |
| caution       | 0.442 | disagreement | 0.387 |
| gloom         | 0.437 | volatility   | 0.384 |
| uncertain     | 0.433 | tension      | 0.383 |
| sensitivity   | 0.427 | certainty    | 0.382 |
| angst         | 0.426 | skepticism   | 0.38  |

# Terms Close to Risk

| term          | sim   | term        | sim   |
|---------------|-------|-------------|-------|
| risks         | 0.737 | misdirected | 0.385 |
| threat        | 0.609 | odds        | 0.379 |
| danger        | 0.541 | uncertainty | 0.375 |
| dangers       | 0.463 | concern     | 0.371 |
| vulnerability | 0.457 | prospect    | 0.37  |
| chances       | 0.451 | instability | 0.363 |
| breakout      | 0.433 | potentially | 0.352 |
| probability   | 0.426 | concerns    | 0.352 |
| possibility   | 0.409 | challenges  | 0.346 |
| likelihood    | 0.406 | risking     | 0.342 |

# Limitations

Typical way of representing documents given a word embedding model is the average embedding for all words in document, e.g. [Gennaro and Ash, 2022].

Word embedding models solve the synonymy problem but not the polysemy or word-order problems.

**Key question** how to obtain embeddings that vary with context?

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# Word Prediction in Large Language Models

To simplify notation, consider sequence of words  $\mathbf{w} = (w_1, \dots, w_N)$ .

The prediction target of **autoregressive** or **generative** language models (e.g., GPT family) is  $w_N \mid \mathbf{w}_{-N}$ .

In **bidirectional** models (e.g., BERT), the prediction target is  $w_n \mid \mathbf{w}_{-n}$ .

In both cases, the prediction is informed by surrounding context.

# Example of Next-Word Prediction Problem

*After a season of positive corporate earnings announcements driven by AI adoption, NVIDIA's share price hit an all-time [MASK].*

Which words are most likely to underlie [MASK]?



## Two Alternative Examples with Six Words Removed

*After a season of ~~positive~~ corporate earnings announcements driven by AI adoption, ~~NVIDIA's~~ share price hit an all-time [MASK].*

*After a season of positive corporate earnings announcements driven by AI adoption, NVIDIA's share price hit an all-time [MASK].*

# Formalizing the Prediction Problem

Endow the masked word  $n$  with an embedding vector  $\rho_n \in \mathbb{R}^K$ .

$\rho_n$  can be used to fit a probability distribution over the  $V$  vocabulary terms that can populate the  $n$ th element of the sequence.

Multinomial regression / feedforward neural network with  $\rho_n$  as input.

How can  $\rho_n$  be built to reflect relevant part of the context?

Also important for construction to be computationally efficient.

# Attention Weights

The basic idea of attention [Vaswani et al., 2017] is to define a weight  $\alpha_{n,m}$  for each **attended** word  $n$  and **context** word  $m$ .

Normalized so that  $\sum_m \alpha_{n,m} = 1$ .

Attention weights highlight the relevant parts of the context surrounding each word.

Weights are estimated during neural network training to optimize the quality of word prediction tasks.

# Parameterization of Attention

Let  $\mathbf{W}_q \in \mathbb{R}^{R \times K}$  and  $\mathbf{W}_k \in \mathbb{R}^{R \times K}$  be **query** and **key** weight matrices.

Steps to generate attention weight  $\alpha_{n,m}$ :

1. Form query vector  $\mathbf{q}_n = \mathbf{W}_q \boldsymbol{\rho}_n$
2. Form key vector  $\mathbf{k}_m = \mathbf{W}_k \boldsymbol{\rho}_m$
3. Compute score  $\tilde{\alpha}_{n,m} = \frac{\mathbf{q}_n \cdot \mathbf{k}_m}{\sqrt{R}}$
4.  $\alpha_{n,m} = \frac{\exp(\tilde{\alpha}_{n,m})}{\sum_{m'} \exp(\tilde{\alpha}_{n,m'})}$

# Using Attention to Update Embeddings

Suppose we have an initial embedding representation  $\rho_n^{(i)}$  for word  $n$ .

Let  $\mathbf{W}_v \in \mathbb{R}^{S \times K}$  be matrix of **value** weights.

Project embedding into value vector  $\mathbf{v}_n^{(i)} = \mathbf{W}_v \rho_n^{(i)}$ .

We obtain a new representation for word  $n$  via

$$\rho_n^{(i+1)} = \mathbf{W}_0 \sum_m \alpha_{n,m} \mathbf{v}_m^{(i)}$$

where  $\mathbf{W}_0 \in \mathbb{R}^{K \times S}$ .

# Multi-Head Attention

Words may be in relationship in many ways.

Different attention weights are typically applied in parallel to update initial embeddings.

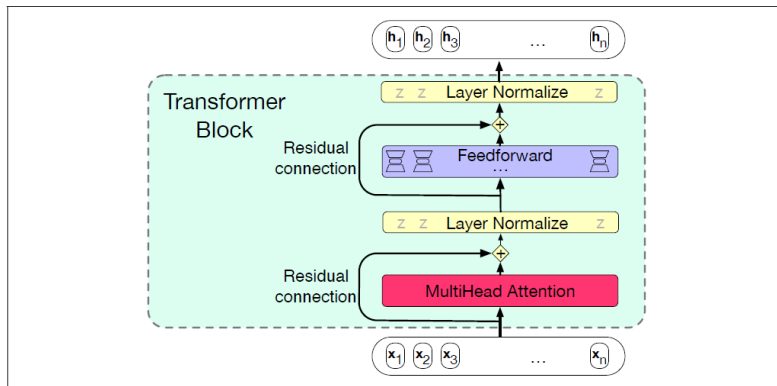
The full operation is called **multi-head attention**.

Each attention operation  $h = 1, \dots, H$  has its own weight matrices.

Fully updated embedding vector is

$$\rho_n^{(i+1)} = \sum_{h=1}^H \mathbf{w}_{0,h} \sum_{m=1}^N \alpha_{n,m}^h \mathbf{v}_m^{(i),h}$$

# Full Transformer Block



**Figure 10.6** A transformer block showing all the layers.

# Initial Embeddings

Transformer blocks take as inputs a sequence of embeddings and output an updated sequence of embeddings.

But so far we have not specified where the initial embeddings come from.

Each word in the input sequence has an initial embedding vector that is the sum of two distinct embeddings:

1. An **embedding for the vocabulary term**.
2. A **positional embedding** that depends on the location of the word in the sequence.

These embeddings are additional estimated network parameters.

Given the estimated structure of the whole network, every input sequence can be processed even if it was not seen in training data.



# Summary of Structure of Large Language Model

1. Begin with input sequence  $w_1, \dots, w_N$ .
2. Assign initial embeddings to each element of sequence.
3. Repeatedly perform the following operations:
  - 3.1 Linearly combine embeddings with attention weights.
  - 3.2 Non-linearly transform each embedding with feed-forward neural network.
4. Output final embeddings for each element of sequence.
5. Use final embeddings for language prediction problem.

# Modeling Choices

While this basic pipeline describes nearly every LLM, there is variety in:

1. Prediction target (e.g. bidirectional vs. autoregressive)
2. Training data
3. Length of context window
4. Number of Transformer blocks
5. Dimensionality of embedding vectors

# BERT

Important example is BERT (Bidirectional Encoding Representations from Transformers) [Devlin et al., 2019].

Trained on BooksCorpus (800M words) and English Wikipedia (2,500M words).

**Masked language modeling.** 15% of words randomly masked and given [MASK] token. [MASK] token embeddings built to successfully predict underlying word.

Original paper had next-sentence prediction but has since been dropped from loss function in extensions [Liu et al., 2019].

Base model has twelve layers, 768-dimensional embeddings, 110M parameters.

# Which Corpus?

Much of traditional text-as-data analysis fits models on corpora drawn from domain of interest.

Large language models were first fit on generic corpora like Common Crawl, Wikipedia, or Google Books.

More recent iterations expand the training data (but details becoming more obscure).

Important to realize that **the training data contains the knowledge that a model can encode.**

Any biases in the training data can also be inherited by the model.

# Example GPT-3 Output

Prompt GPT-3 [Brown et al., 2020] with He was very [MASK] and She was very [MASK].

**Table 6.1:** Most Biased Descriptive Words in 175B Model

| Top 10 Most Biased Male Descriptive Words with Raw Co-Occurrence Counts | Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts |
|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Average Number of Co-Occurrences Across All Words: 17.5                 | Average Number of Co-Occurrences Across All Words: 23.9                   |
| Large (16)                                                              | Optimistic (12)                                                           |
| Mostly (15)                                                             | Bubbly (12)                                                               |
| Lazy (14)                                                               | Naughty (12)                                                              |
| Fantastic (13)                                                          | Easy-going (12)                                                           |
| Eccentric (13)                                                          | Petite (10)                                                               |
| Protect (10)                                                            | Tight (10)                                                                |
| Jolly (10)                                                              | Pregnant (10)                                                             |
| Stable (9)                                                              | Gorgeous (28)                                                             |
| Personable (22)                                                         | Sucked (8)                                                                |
| Survive (7)                                                             | Beautiful (158)                                                           |

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# Distance between Documents

Both bidirectional and autoregressive language models produce vector representations of word sequences.

Cosine similarity between these representations can then be used to measure distance.

Plausible argument for preferring bidirectional models, which exploit full structure of document.

Unlike previous methods, LLMs can produce different vectors for  
*economic growth is weak but long-term productivity trends are strong*  
*economic growth is strong but long-term productivity trends are weak*

See [Jha et al., 2022] for example.

# Concept Detection

The structure of autoregressive language models suggests their use for concept detection.

Formulate a sequence of words such as

*“Consider the sentence ‘the economy is booming’. Is the sentiment in this sentence positive, neutral, or negative?”*

This text is converted to  $\mathbf{w} = (w_1, \dots, w_N)$ .

$N + 1$ th word is drawn from  $\Pr[w_{N+1} \mid \mathbf{w}]$ .

$N + 2$ th word is drawn from  $\Pr[w_{N+2} \mid \mathbf{w}, w_{N+1}]$ .

And so forth.

The ability to generate seemingly correct responses is called **zero-shot learning**.



# Temperature

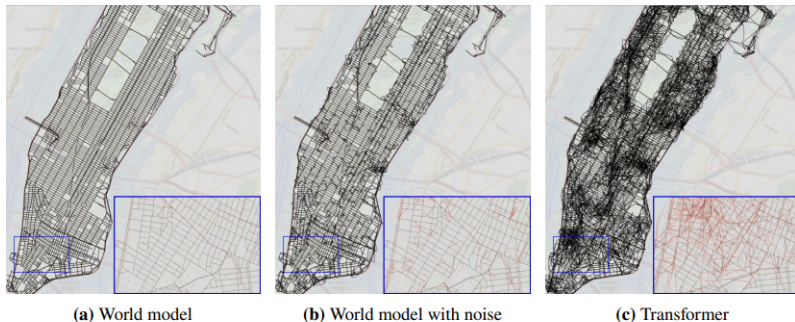
One important modeling choice is how exactly to sample from  $\Pr[w_{N+1} \mid \mathbf{w}]$ .

Always choosing most likely word can make response rigid.

Sampling from the full distribution allows more creativity but can produce less accurate responses.

The **temperature** is a parameter that controls the trade-off between quality and diversity.

# Accurate Prediction $\neq$ True Model



**Figure 3:** Reconstructed maps of Manhattan from sequences produced by three models: the true world model (left), the true world model corrupted with noise (middle), and a transformer trained on random walks (right). Edges exit nodes in their specified cardinal direction. In the zoomed-in images, edges belonging to the true graph are black and false edges added by the reconstruction algorithm are red. We host interactive reconstructed maps from transformers at the following links: [shortest paths](#), [noisy shortest paths](#), and [random walks](#).

From [Vafa et al., 2024].

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